

(Treatment of genetic diseases)

Gene therapy:

-Gene therapy techniques involve the replacement of a non-functional or missing gene product by inserting normal gene into a somatic cell (Hepatocyte, Lymphocyte, Skin fibroblast.

Endothelial cell brain cell). There are many physical, chemical, and biological methods being used for introducing genes in to cells, physical methods include microinjection, electroporation, partical bombarment and chemical methods include liposomes that carry DNA across the plasma membrane.

Biological approaches to gene transfer use a vector, such as viral genome.

- Some viruses such as **Retroviruses** and **Adenoviruses** are able to insert copies of normal human gene into the affected cells.

- Transplantation of cells, tissue or organs can be considered as a form gene therapy because the cells transferred to the recipient continue to function like the donor genome, ex: Liver transplantation, Bone marrow transplantation.

- Replacement of deficient substance may be a possible mode of treatment, ex: in classical Hemophilia replacement of the deficient clotting factors VIII correct the bleeding disorder.
- The type I insulin-dependent diabetes mellitus is treated by administration of insulin.

There are two main types of gene therapy : 1-somatic cell gene therapy and 2-reproductive or germ-line gene therapy.

Germ-line cell therapy involves the introduction of corrective genes into reproductive cells (sperm and eggs) or zygotes, with the objective of creating a beneficial genetic change that is transmitted to the offspring.

Gene therapy of somatic cells, those not directly related to reproduction, results in changes that are not transmitted to offspring. An example of gene therapy in somatic cells is the introduction of ADA gene into an organ or tissue (bone marrow) to induce the production of an enzyme the ADA (adenosine deaminase) for patient with Severe combined immunodeficiency disease.